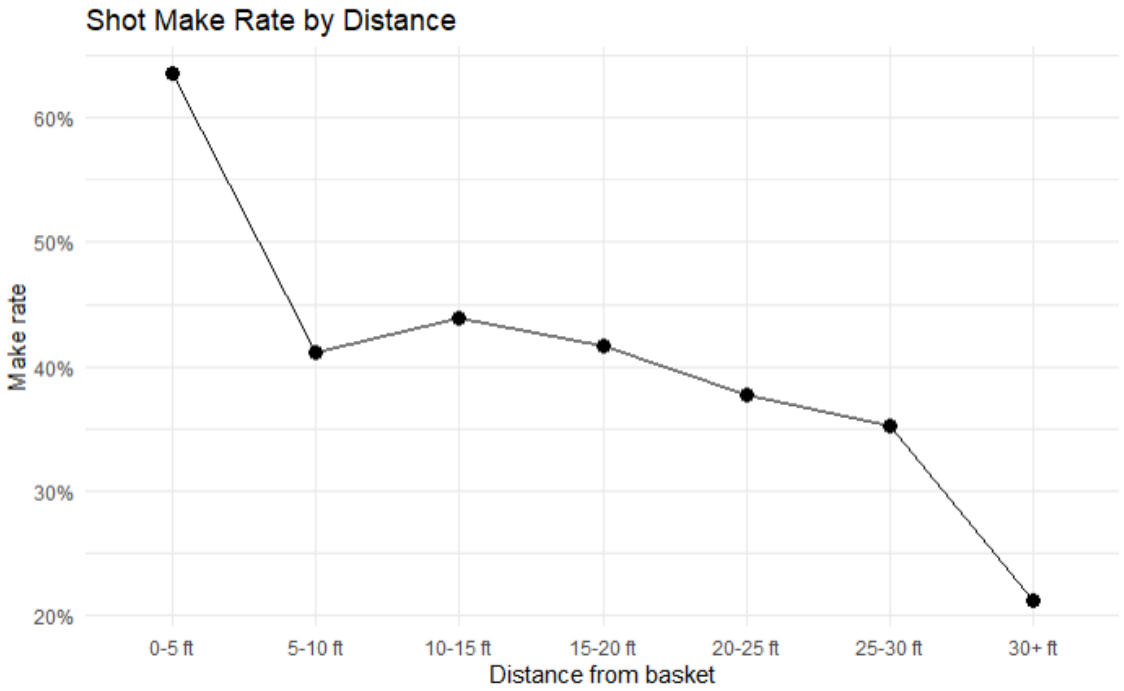
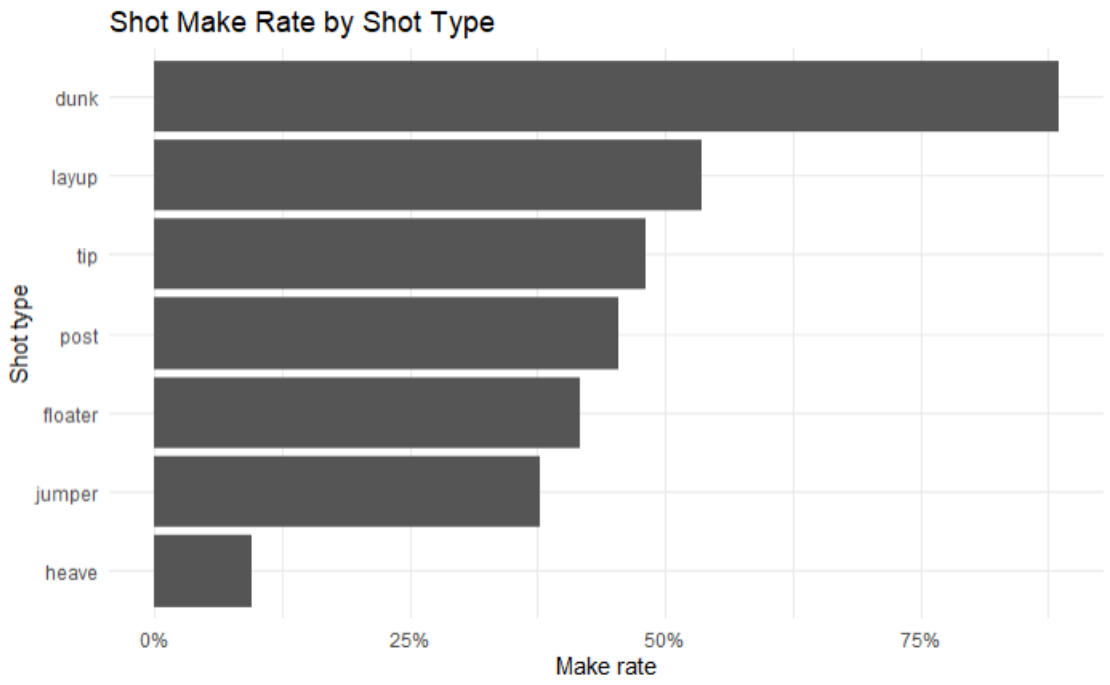
Exploratory Data Analysis

Beginning this project, I examined the distribution of the response variable and their relationships between shot outcomes. This required the other predictors that were provided in the data. It was clear that 46% of the shots in the training data were made and 54% were misses.

Shot distance was an easy variable to identify that had a relationship with shooting success. I grouped shot attempts into distance intervals so that shot distance can be categorized into groups. It is clear that the make rate decreases as distance from the basket increases. I understood that distance would be an important predictor and that its relationship with shot probability may not be linear.



Shot types showed differences in observed make rates. For example, dunks had a much higher make rate than the other shot types. I understood that shot type can be included as a categorical predictor instead of trying to use distance alone.



Defensive pressure was another factor I looked into. I grouped distance between shooter and closest defenders similar to shot distance. The observed make rates differed across the number of contestants.

Finally, I used shot location to build a two-dimensional summary of shot outcomes. This showed shooting efficiency varying across the court and that nonlinear relationships may be present between the predictors.

Baseline and Featuring Engineering

I created an 80/20 training-validation split with a fixed random seed. I kept this validation set throughout the model development so that models can stay consistent. My baseline model used **shot distance, two- versus three-point status, shot type, closest defender distance, game state, shot clock, dribbles before the shot, shooter speed, X court location, Y court location**. The baseline logistic regression achieved a validation log-loss of **0.6447523.**

I then wanted to see if the number of contestants could have an effect so I added it to the nonlinear logistic regression which further reduced validation log-loss to **0.64033**.

In addition, I engineered a defender approaching feature. This variable contained four groups of defender distance. I got those measurements and calculated the change between the defenders

distance one second before the attempt and 0.25 seconds before the attempt. The negative values reveals the defender moving closer to the shooter.
I realized that this feature didn't add much to the logistic model's validation log-loss which is why I left it out of the final model.

XGBoost Model Development

Once I reached my logistic-regression benchmarks, I asked AI to assist me with multiple different routes that I can take my model. It gave me ideas such as Logistic Regression, GAM, Decision Tree, Random Forest, and XGBoost. I choose XGBoost because many relationships in shot prediction involve nonlinear interactions. My initial XGBoost model used the primary shot and defensive variables and got a validation log-loss of **0.6278**.

I then compared tree depths of 3,4,5, and 6.

Maximum Depth	Best Iteration	Validation Log-Loss
3	600	0.6285151
4	425	0.6279825
5	273	0.6276928
6	217	0.6275743

The maximum tree depth 6 produced the lowest validation log-loss.

After this, I checked whether the individual defender-distance variables could be useful information. Distcont4 was excluded because the majority of the data was null. I added the other contested distance variables and improved validation to 0.62718.

Hyperparameter Tuning and Model Selection

After selecting a tree depth of 6 and adding defensive features, I tuned the model with learning rate (eta) and min child weight.

Eta	Min Child Weight	Best Iteration	Validation Log-Loss
0.03	5	321	0.6269934
0.03	**10**	**359**	**0.6269139**
0.05	5	268	0.6271396
0.05	10	206	0.6270708

The best validation result was eta = 0.03 and min child weight = 10. This reached a validation log-loss of 0.6269139 at 359 boosting iterations.

This is the overall progression of my model development:
The largest improvement came from shifting from logistic regression to XGBoost model.

Model	Validation Log-Loss
Baseline logistic regression	0.6447523
Nonlinear logistic regression	0.6424506
Nonlinear logistic + contest information	~0.64033
Initial XGBoost	~0.6278
XGBoost, depth 6	0.6275743
XGBoost + expanded defender distances	~0.62718
Final tuned XGBoost	**0.6269139**

Model Interpretation

XGBoost model revealed that shot distance was the dominant predictor while variables such as closest defender distance and number of contestants were also strong predictors. Shot type,

shooter speed, shot clock, and court location also contributed to the model. XGBoost made it so that variables such as court location could be interacted in a nonlinear way.

Limitations and Problems

A limitation this project had was the hyperparameter tuning as I limited it so that it wouldn't be an exhaustive search. I focused on a small number of parameters that seemed to be most relevant after creating a baseline. With more time, I would try to follow a systematic tuning procedure instead of limiting its searches. I would like to use model-interpretation techniques to fully understand the nonlinear relationships and interactions throughout the model.

One of the problems I ran into was handling the defensive tracking variables. A lot of these variables contained missing values but it was not a simple removal because then the dataset would be missing a lot of values. When I first tried to include defender distance, my dataset lost a lot of its data so I had to make sure that I specified the missing values directly. This way, XGBoost can handle them rather than removing the observation.

Another problem I had was working with the closest defender approach data. This contained multiple defender distance measurements within an observation. To work around this I separated the measurements and created a new defender_approach feature. This highlighted the difference between an observed association and a casual effect. Once I added it to the model, it didn't produce much of an improvement so I excluded it from the final model.

Overall, my final approach combined exploratory analysis, logistic regression benchmarks, targeted engineering, and limited hyperparameter tuning to reduce the validation log-loss from **0.6447523** to **0.6269139**. The analysis provides an understanding of variables such as shot distance and how they provide predictive features. I enjoyed working with real-time basketball data and applying my skills that I have gained over the years of learning about data science to build a predictive model.